

TANGSHAN, China

WMO: 545340

Lat: 39.5816N

Lon: 118.0931E

Elev: 95

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 7.3	(c) 11.2	(d) -15.8	(e) 2.3	(f) 21.6	(g) -12.0	(h) 2.9	(i) 20.9	(j) 20.8	(k) 28.6	(l) 18.1	(m) 26.2	(n) 3.4	(o) 320	(p) 0.422

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	92.9	73.8	90.5	73.8	88.3	73.3	80.7	87.5	79.1	85.4	77.5	83.7	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	149.9	85.3	77.2	142.1	83.5	75.7	134.9	82.0	44.6	87.7	42.8	85.6	41.3	83.9	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.3	14.3	12.1		3.3	97.5	6.4	3.1	-1.3	99.7	-5.1	101.5	-8.7	103.2	-13.3	105.4	
			DB	1.5	82.8	5.6	1.5	-2.5	83.9	-5.8	84.7	-9.0	85.6	-13.0	86.6	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	24.5	31.2	43.6	56.9	67.9	75.7	80.2	78.1	69.9	56.5	40.8	28.3	(6)
	DBStd	20.37	6.35	6.84	7.38	6.93	5.81	4.84	3.96	3.95	5.58	6.84	7.60	6.08	(7)
	HDD50	2541	789	527	225	18	0	0	0	0	0	22	288	672	(8)
	HDD65	5312	1254	947	662	254	36	1	0	0	21	273	727	1137	(9)
	CDD50	4217	0	0	27	226	556	771	935	872	596	223	11	0	(10)
	CDD65	1514	0	0	0	12	127	322	470	407	167	9	0	0	(11)
	CDH74	13839	0	0	4	127	1169	2903	4767	3700	1114	55	0	0	(12)
	CDH80	4894	0	0	0	16	339	1118	1895	1263	257	6	0	0	(13)
	WSAvg	5.4	5.0	5.4	6.5	7.0	6.7	5.8	5.0	4.4	4.5	4.6	4.7	4.7	(14)
	PrecAvg	25.0	0.2	0.2	0.4	1.0	1.5	3.3	7.7	7.2	2.2	0.9	0.3	0.2	(15)
Precipitation	PrecMax	39.7	1.0	1.2	1.5	3.6	4.3	7.8	15.6	13.8	6.6	2.7	1.3	1.0	(16)
	PrecMin	15.2	0.0	0.0	0.0	0.0	0.2	0.7	1.1	1.9	0.1	0.0	0.0	0.0	(17)
	PrecStd	7.0	0.2	0.3	0.4	0.9	1.0	1.9	3.6	3.4	1.5	0.7	0.3	0.2	(18)
	PrecStd	7.0	0.2	0.3	0.4	0.9	1.0	1.9	3.6	3.4	1.5	0.7	0.3	0.2	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	54.3	71.5	81.9	91.8	96.5	97.1	93.2	88.3	79.8	64.1	47.5	(19)
		MCWB	33.7	40.6	52.8	59.8	65.7	68.6	74.0	78.2	70.3	62.4	50.0	36.8	(20)
	2%	DB	40.6	49.8	65.8	77.5	87.3	92.1	93.5	90.3	85.4	74.7	60.0	43.6	(21)
		MCWB	31.1	38.1	49.2	58.0	64.0	69.7	75.9	76.4	68.5	60.0	48.5	35.5	(22)
	5%	DB	37.6	46.3	60.7	73.9	83.8	89.3	90.9	88.2	82.9	71.0	56.1	40.5	(23)
		MCWB	29.3	36.3	46.2	55.8	63.1	69.5	76.3	75.6	68.0	58.1	46.5	33.1	(24)
	10%	DB	34.8	42.9	56.4	70.2	80.7	86.3	88.6	86.3	80.4	67.9	52.5	37.8	(25)
		MCWB	28.0	34.6	43.2	54.0	61.8	68.8	75.8	74.8	66.7	56.2	44.4	31.5	(26)
	0.4%	WB	34.8	42.8	55.8	64.6	71.1	76.6	82.8	82.6	77.0	66.6	54.8	40.8	(27)
		MCDB	41.9	49.9	68.1	77.5	85.9	87.1	90.4	88.7	82.5	73.7	59.4	44.3	(28)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.2	39.9	51.0	61.0	68.4	74.4	81.0	80.7	73.4	64.1	50.8	36.7	(29)
		MCDB	38.2	46.9	62.1	73.3	81.5	83.9	88.4	86.6	80.2	70.3	56.4	41.0	(30)
	5%	WB	30.5	37.4	47.4	58.5	66.3	72.9	79.7	79.0	71.5	61.5	48.1	34.4	(31)
		MCDB	36.4	45.1	58.3	69.6	78.3	82.4	86.7	84.7	77.8	67.7	54.5	39.0	(32)
	10%	WB	28.7	34.9	44.5	56.0	64.4	71.4	78.2	77.5	69.7	59.1	45.5	32.3	(33)
		MCDB	34.3	42.3	55.2	66.7	76.1	81.4	84.8	83.0	76.5	65.1	51.5	37.0	(34)
Mean Daily Temperature Range	5% DB	MDBR	17.6	18.1	19.2	20.0	20.4	17.3	13.9	14.5	17.8	18.9	17.0	16.1	(35)
		MCDBR	20.5	21.9	25.1	25.3	26.2	22.7	17.7	16.7	20.4	22.5	21.2	19.6	(36)
	5% WB	MCWBR	13.4	13.3	12.7	11.2	9.7	7.4	6.2	6.4	7.4	9.9	11.6	12.7	(37)
		MCWBR	18.6	19.9	23.2	21.1	20.8	16.6	13.9	13.3	15.4	17.7	19.1	16.8	(38)
	5% WB	MCWBR	12.5	12.8	12.4	11.5	9.6	7.2	6.4	6.6	7.7	9.3	11.6	11.6	(39)
		MCWBR	12.5	12.8	12.4	11.5	9.6	7.2	6.4	6.6	7.7	9.3	11.6	11.6	(39)
Clear-Sky Solar Irradiance	taub		0.439	0.519	0.545	0.621	0.666	0.789	0.792	0.729	0.655	0.631	0.508	0.457	(40)
	taud		1.922	1.779	1.750	1.649	1.600	1.448	1.482	1.576	1.650	1.642	1.832	1.923	(41)
	Ebn at Noon		212	211	223	217	211	186	184	191	193	176	186	193	(42)
	Edh at Noon		43	58	66	77	83	97	93	82	72	66	48	41	(43)
All-Sky Solar Radiation	RadAvg		756	990	1425	1698	1904	1686	1519	1490	1339	1022	767	661	(44)
	RadStd		74	103	122	84	118	161	119	89	100	93	97	63	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+83

Nomenclature: See separate page